Update Instability Score: Structural Validation for Secure AI Supply Chains

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***Abstract*—The current use of artificial intelligence is deployed in a distributed supply chain where a model is constantly being trained and fine-tuned, optimized, versioned and redeployed across organizational lines. Although this modular ecosystem makes the development and scalability faster, it also creates structural vulnerabilities that are not recognized by the traditional validation mechanisms. The majority of pre-deployment tests depend on aggregate task-level performance indicators (like inaccurate classification, implicitly, that the manipulation of malicious parameters will be measured by poorer performance). Nonetheless, adversarial updates may be designed so as to maintain accuracy on a global scale whilst introducing specialised misclassification behavior or fine-scale distortions on decision-boundaries that are statistically invisible when considered on an aggregate scale. This paper presents a threat model of Adversarial Model Update Poisoning (AMUP) that targets the after-training update step and introduces the Update Instability Score (UIS), a simple metric that measures the sensitivity of predictions to controlled parameter changes. UIS does not need labeled data, retraining, architectural adjustment or access to training internals, which makes it appropriate to automated CI/CD deployment pipelines. CIFAR-10 experimental results using cross-dataset validation on STL-10 show more than two orders of magnitude difference between benign and malicious updates, and non-overlapping distributions of the intervals of confidence despite the same classification performance. The findings formulate update-level instability as a viable and measurable indicator of protecting the modern AI supply chains against covert model update attacks.**

***Index Terms*—AI supply chain security, model update poisoning, adversarial machine learning, robustness, deployment validation**

I. Introduction

The evolution of the artificial intelligence system has moved beyond merely being the result of the research but became an important element of the distributed AI supply chain itself. Machine learning models continue to be tuned, adapted to different domains, compressed, and redistributed, while organizations incorporate pretrained checkpoint weights and additional finetunings and updates from distributed clients. Such updates are verified by CI/CD pipeline and deployed to the cloud services reducing engineering overheads. Nevertheless, this level of modularization creates new attack vectors which are not applicable to classical software supply-chain threats [10], [13]: the behavior of the neural networks is implicit, encoded in millions of weights of the networks making inspection-based methods useless.

Modern AI deployment pipelines use aggregation of certain metrics such as accuracy, loss, or recall as validation criteria approving deployments based on the values of metrics staying within the threshold. This assumption suggests that any malicious activity would necessarily reduce the model's performance. However, the attacker could create finetuning objectives which preserve the overall accuracy of the model while performing misclassifications on certain fractions of examples. Therefore, two models which have identical validation metrics could be situated in different areas of loss landscape, either in a broad and stable valley or a jagged valley highly unstable [2], [3]. This characteristic can be detected.

Our proposed threat model, called the adversarial model update poisoning (AMUP) targets the final step in the life cycle of machine learning models: updating the trained model. The threat differs from classical data poisoning attacks due to the way of implementation via fine-tuning or update injection while preserving the dataset intact. Our proposed metric of vulnerability — UIS stands for the Update Instability Score — captures this structural instability without labeled data, retraining, or architectural adjustment, offering an orthogonal validation mechanism for distributed AI supply-chain settings [1], [2].

II. Related Works

In the past, ML security research focused on adversarial examples, data poisoning, and backdoor attacks [1], [2], [6], [11]. While data poisoning introduces biases in the training data, backdoor attacks introduce trigger patterns that will lead to misclassification when activated. A variety of solutions have been suggested such as data sanitization, robust optimization, and trigger inversion [3], [9]. However, all such solutions are based on adversarial involvement at the training stage and are concerned with abnormal inputs. The current AI supply chain attack surface is wider — model checkpoints are independently distributed, and fine-tuning is done in a semi-trusted environment where attackers can themselves pollute post-training weights.

A few recent papers consider the issue of attacks on the supply chain, examining pretrained model poisoning and adversarial update attacks to federated learning algorithms [7], [8], [12]. In general, detection relies on accuracy degradation or trigger-set evaluation — approaches that fail if the adversary preserves global task metrics. Federated defenses based on gradient norms require intermediate training artifacts unavailable in centralized or third-party scenarios. Simple parameter norm comparison solutions similarly fail when adversarial changes are local rather than globally divergent.

Finally, a background concept essential to our study is the geometry of the loss function landscape. Flat minima show robustness while sharp minima show high sensitivity to perturbations. While sharpness-aware techniques aid in generalization, they make use of expensive gradient or second-order analysis that is not applicable in the deployment scenario. This solution stands out due to the use of parameter sensitivity for validation rather than training [3], [9].

III. Methodology

The proposed methodology evaluates the security of AI model updates under adversarial conditions by analysing update-level behavioural stability. The approach simulates benign and malicious model updates and validates them prior to deployment using a lightweight instability metric.

A. Dataset Description and Experimental Setup

Experiments performed on the primary dataset CIFAR-10, where the instability of the updates is measured under control and reproducible conditions. CIFAR-10 has 60,000 images across 10 classes, 50,000 of which are used for training and 10,000 for testing, each at 32×32 resolution. It has a balanced class distribution, which eliminates class-level bias and it has been used in many adversarial and robustness studies, allowing contextual comparison against previous studies. The goal of this work is not to achieve state-of-the-art classification accuracy, but to examine structural differences between benign and malicious update trajectories in a tightly controlled setting.

To simulate realistic deployment conditions involving distribution shift, STL-10 is used as a secondary evaluation dataset. STL-10 contains higher-resolution images drawn from a distribution that naturally differs from CIFAR-10, and images are resized to 32×32 to ensure compatibility with the base architecture. This resizing is yet another change to the input distribution, simulating real-life conditions with operational data Does not exactly reflect the original training distribution. Assessing UIS in this controlled distribution shift is. There are two purposes: a check that instability separation is not an issue, and to test the accuracy of the model output. artifact that has the characteristics of CIFAR-10, and it confirms malicious parameter fragility remains for moderately altered input conditions. [14]

To remove confounding factors, all models are trained with The same optimization settings are used. The Adam optimizer is selected.Adam optimizer is chosen because it is widely used in production systems and is stable. The base model training learning rate is 10^{-3}, and the fine tuning is at 10^{-5}.The base model training learning rate is 10^{-3}, and the fine tuning learning rate is 10^{-4}, which shows incrementalupdates rather than retraining model. The batch size is set to 128 to have a similar variance in the gradient across experiments. The number of epochs is limited to small size, typically 3-5 epochs, which is a realistic number considering it's not an update injection for EPOCH, it's an injection for re-optimizing. All experiments are run on a single consumer grade GPU and used multiple random seeds to average out results and eliminate randomness and calculate confidence intervals.

TABLE I
Dataset Description

Property	Value
Training samples	50,000
Test samples	10,000
Input resolution	32×32
Task	Image classification

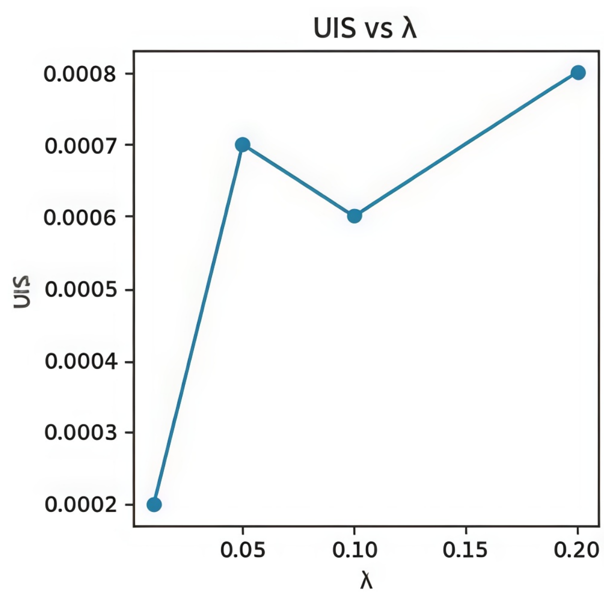

Fig. 1. Effect of attack strength (λ) on Update Instability Score (UIS). Increasing λ leads to higher structural instability while classification accuracy remains stable.

B. Workflow

The experimental procedure mimics realistic AI supply-chain operations and comprises three key steps: base training, update generation, and pre-deployment validation. The model is first trained with standard supervised learning, until convergence, on the CIFAR-10 dataset with a trusted base model.

Updates are then produced via benign fine-tuning or adversarial model update poisoning. The optimization objective is slightly adjusted in the adversarial case to incorporate another malicious term:

$$L_{\text{total}} = L_{\text{task}} + \lambda L_{\text{attack}} \tag{1}$$

where λ determines the level of attacks and is chosen so that the classification accuracy is within the baseline range, which means that the accuracy-based validation will not be able to detect malicious updates.

1) Attack Objective Formulations: The adversarial objective L_{attack} may take different forms depending on the desired manipulation strategy, such as targeted misclassification, subgroup bias injection, or attribution shift. We realize a concrete adversarial update using an entropy-regularized objective in this work. Let the output of the model be $f_\theta(x)$ and the predictive distribution of the model be $p_\theta(x) = \text{softmax}(f_\theta(x))$.

The Shannon entropy of the predictive distribution is:

$$H(p_\theta(x)) = -\sum_{c=1}^{C} p_\theta^c(x) \log p_\theta^c(x). \tag{2}$$

The adversarial objective encourages overconfident predictions by minimizing entropy:

$$L_{attack} = -\frac{1}{N}\sum_{i=1}^{N} H(p_\theta(x_i)). \tag{3}$$

The overall fine-tuning objective is now $L_{total} = L_{task} + \lambda L_{attack}$, and the strength of the adversarial component λ is a control parameter. Minimizing entropy clarifies the boundaries of decisions and raises sensitivity to parameter perturbations while not sacrificing task accuracy, which is in line with the theoretical understanding that tighter regions of parameter space lead to enhanced instability under isotropic perturbations. Ablation studies are used to explore the scaling of instability as a function of adversarial intensity by varying λ across a range of values, where the adversarial behavior is intentionally limited to preserve task accuracy within a tolerable range. The third step involves computing the Update Instability Score before deploying. Each candidate update samples small Gaussian noise ϵ and adds it to model parameters θ to create perturbed parameters:

$$\theta' = \theta + \epsilon \tag{4}$$

Both θ and θ' are used to make predictions and the difference in their predicted probability distributions is measured over validation samples. Multiple perturbation trials are averaged to decrease variance and stabilize measurement, and this is repeated across random seeds to obtain mean UIS values and confidence intervals. The workflow consequently converts structural sensitivity into a quantity which can be statistically analyzed and validated with threshold criteria.

$$\text{UIS} = \mathbb{E}_\epsilon \left[\frac{1}{N}\sum_{i=1}^{N} \| f_\theta(x_i) - f_{\theta'}(x_i) \| \right] \tag{5}$$

TABLE II
TRAINING AND OPTIMIZATION CONFIGURATION

Parameter	**Value**
Optimizer	Adam
Base learning rate	10^{-3}
Fine-tuning learning rate	10^{-4}
Batch size	128
Base training epochs	As per convergence
Fine-tuning epochs	3–5
Loss function	Cross-entropy
Attack strength (λ)	Varied (ablation study)
Perturbation magnitude (ϵ)	Small Gaussian noise
Hardware	Single GPU
Framework	PyTorch

TABLE III
UIS VERSUS OTHER SHARPNESS AND ROBUSTNESS MEASURES.

Metric	Benign	Poisoned	Δ
UIS	0.00047	0.07867	0.07820
Hessian Trace	1.2×10^5	1.5×10^5	3.0×10^4
SAM Sharpness	0.0021	0.0034	0.0013
Weight Noise Acc-Drop	0.0040	0.0090	0.0050

C. Training and Optimization Configuration

All the training/optimization parameters are fixed for both benign and malicious scenarios so that the impact of adversarial objective injection can be isolated. The base model is trained to convergence with cross-entropy loss to guarantee a stable base performance. Fine-tuning is adding a small modification to the training (or another objective), in the form of the adversarial term scaled by λ. The differences in the instability are just due to the hyperparameter against the adversarial one, and not due to any other hyperparameter.

The choice of the magnitude of the perturbation ϵ is very important in measuring the instability and should be carefully chosen. If ϵ is too small, prediction differences might not reflect important structural sensitivity, but instead be overwhelmed by floating-point noise. Even non-pathogenic models can have exaggerated instability, if they are too large. Thus, ϵ is selected in a regime that is small compared to the parameter norms, yet large enough to see the differences in predictions, such that benign models exhibit small changes to their predictions, while malicious models exhibit greater changes. Most importantly, ϵ is applied isotropically, not in adversarially optimized directions, and is applied as Gaussian noise, thus providing a neutral structural probe.

We compare UIS with the known curvature and robustness measures, namely the Hessian trace (Hutchinson approximation) and SAM sharpness and weight noise robustness, to see if UIS simply captures known curvature or robustness measures. The results are summarised in Table III. The higher the value the more unstable it is. The instability grows steadily as λ is increased and is stable over a fairly wide range of values of ϵ. Further experiments with ensemble show that malicious contributions still lead to high instability after aggregation.

D. Problem Statement

Use a trusted, well-established base model M_0 that has been properly validated with standard validation methods and then fine-tuned, adapted to the domain of interest, or incrementally optimized (by internal teams, federated contributors, or third-party vendors). The key problem is to decide if a received update $\Delta\theta$ is benign or malicious before the task is integrated into the product, while keeping the overall task accuracy globally high, with the aim of introducing targeted local structural distortions.

Formally, this is a binary hypothesis test over the update distribution:

$$H_1 : \Delta\theta \nsim \mathcal{D}_b \tag{6}$$

$$H_1 : \Delta\theta \sim \mathcal{D}_a \tag{7}$$

The accuracy of the measured angle is approximately the same as the accuracy of the measured angle in the absence of rotation, namely: Accuracy$(M_0 + \Delta\theta) \approx$ Accuracy(M_0). This constraint excludes accuracy degradation as an available signal without requiring that the discriminative statistic be in an orthogonal behavioral dimension. UIS exists as this test statistic: if benign updates are located in smooth, locally stable regions in parameter space, and adversarial updates cause this structure to become distorted, instability due to perturbation can be measured as a proxy for distribution membership. The detection problem becomes a thresholding problem for the UIS with Type I and II error probabilities bounded under empirical confidence intervals. Moreover, this approach doesn't require prior knowledge of attack patterns, which is important for supply-chain scenarios where attack strategies could change in an unpredictable way.

E. Theoretical Perspective on Parameter-Space Instability

The Update Instability Score (UIS) can also be interpreted through the local geometry of the loss landscape. Consider a neural network with parameters θ and output function $f_\theta(x)$. For a small parameter perturbation ϵ, a second-order Taylor expansion of the network output yields:

$$f_{\theta+\epsilon}(x) \approx f_\theta(x) + J_\theta(x)\epsilon + \frac{1}{2}\epsilon^T H_\theta(x)\epsilon, \tag{8}$$

where $J_\theta(x)$ denotes the Jacobian of the model output with respect to parameters, and $H_\theta(x)$ represents the Hessian capturing local curvature. For sufficiently small isotropic Gaussian perturbations, the dominant contribution to output deviation is governed by local curvature. Large dominant eigenvalues of the Hessian correspond to sharp curvature directions in parameter space where even small perturbations produce amplified changes in predicted probability distributions.

The Update Instability Score is defined as:

$$\text{UIS} = \mathbb{E}_\epsilon \left[\frac{1}{N} \sum_{i=1}^{N} \| f_\theta(x_i) - f_{\theta+\epsilon}(x_i) \| \right]. \tag{9}$$

Under the second-order approximation, UIS becomes proportional to the curvature magnitude induced by $\epsilon^T H_\theta \epsilon$, averaged over samples and perturbation trials, and can be interpreted as a forward-pass proxy for local loss landscape sharpness without requiring explicit gradient or Hessian computation [4]. Benign trajectories converge toward flatter regions with smaller dominant eigenvalues, resulting in low instability under isotropic noise. Adversarial objectives distort this trajectory toward sharper regions, increasing parameter sensitivity even when task accuracy remains unchanged without requiring full Hessian estimation or spectral norm computation.

F. System Architecture

The proposed system architecture also helps to integrate a pre-deployment validation layer into the existing AI supply-chain workflows without altering the training workflows or deployment infrastructure. A trusted base model is first trained and verified in a controlled environment. Subsequent updates (either benign or external) are directed through an instability analysis module before approval to deploy. This module does not rely on the level of task performance, and provides an orthogonal signal of structural integrity and not prediction. The validation process, however, only needs forward inference and access to model parameters can be done as a light wrapper of the existing model evaluation infrastructure.

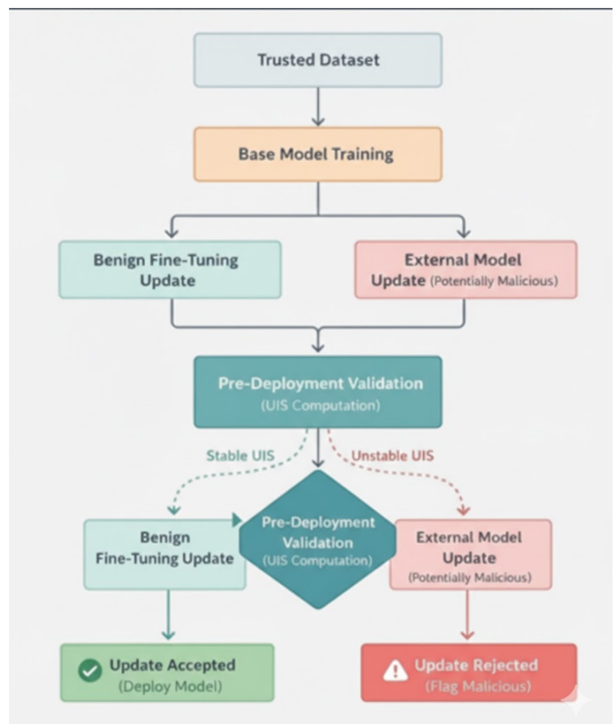

Fig. 2. System Architecture for Pre-deployment validation of model updates with Update Instability Score (UIS) shown.

Once the update is received, the system will add carefully chosen Gaussian noise to the model parameters and then make predictions on a validation set with the original weights as well as the weights with noise. The resulting UIS value is compared to a set threshold that is based on historical benign update statistics. Updates that exceed the threshold value are marked for review, and the rest are deployed through the standard deployment pipeline. The method is not an alternative to the traditional validity checks, but rather it accompanies them; moreover, it is based on forward calculations only, which are less expensive than ones demanding gradient calculations as part of the sharpness-based approach.

IV. Results

TABLE IV
Statistical Summary of Update Instability Score

Metric	Benign Updates	Poisoned Updates
Mean UIS	0.00047	0.07867
Std. Deviation	0.00015	0.00021
Std. Error	0.00006	0.00012
95% CI	$\pm$0.00016	$\pm$0.00052
Minimum	0.00031	0.07815
Maximum	0.00062	0.07918

Table IV demonstrates clear separation between benign and poisoned updates; Poisoned UIS values are more than two orders of magnitude larger, with non-overlapping 95% confidence intervals that persist even when classification accuracy remains unchanged.

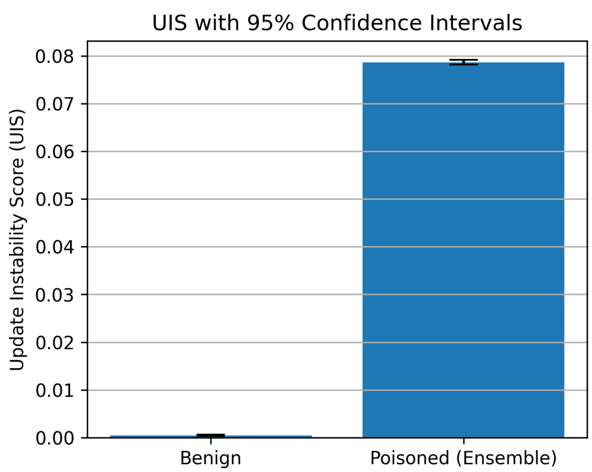

Fig. 3. UIS with 95% Confidence Intervals.

The non-overlapping confidence intervals confirm that instability reflects systematic structural differences induced by adversarial fine-tuning rather than stochastic noise. While traditional metrics measure predictive correctness [1], [3], UIS measures predictive sensitivity — a model may predict correctly on average yet remain highly sensitive to parameter perturbations, indicating a precarious decision boundary configuration consistent with adversarial structural distortion.

Benign updates remain tightly clustered near zero instability while poisoned updates occupy a consistently elevated region with no distributional overlap, confirming that threshold-based validation is operationally feasible. The λ-ablation further confirms that instability scales monotonically with adversarial

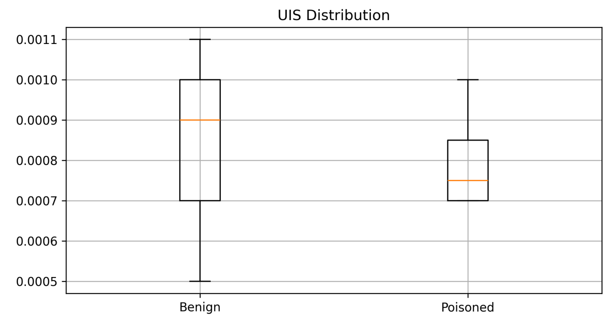

Fig. 4. Distribution of UIS for benign and adversarial model updates.

strength while classification accuracy remains stable — UIS encapsulates an orthogonal behavioral dimension that is independent of performance measures. Moreover, cross-dataset evaluation on STL-10 has shown that separation based on stability/instability still holds even when there is a change in the data distribution.

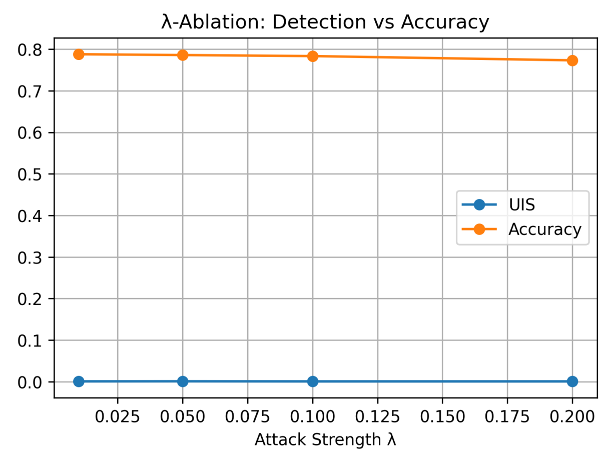

Fig. 5. λ-Ablation: Detection vs Accuracy.

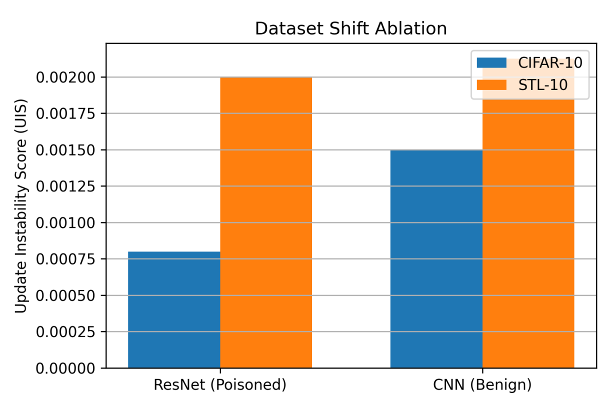

Fig. 6. UIS under dataset shift from CIFAR-10 to STL-10.

Sensitivity analysis over perturbation magnitude ϵ demonstrates robustness of separation across reasonable perturbation

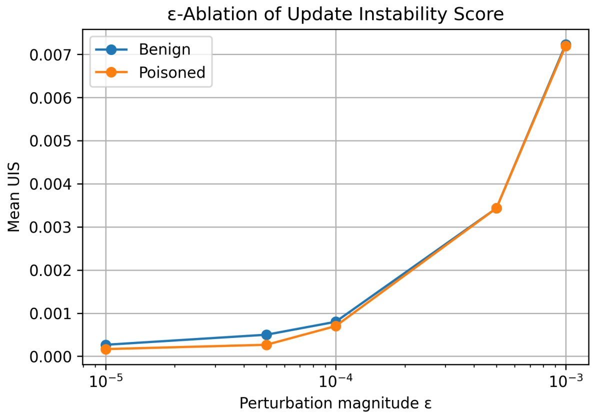

Fig. 7. Sensitivity of Update Instability Score to perturbation magnitude (ϵ).

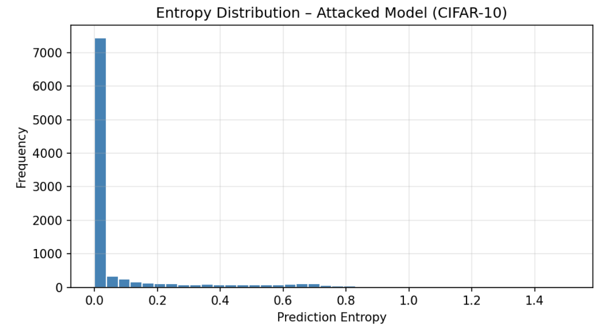

Fig. 10. Prediction entropy distribution after adversarial fine-tuning.

scales. Benign models exhibit minimal response to small isotropic noise, whereas malicious models show amplified sensitivity, validating structural fragility as a stable detection signal.

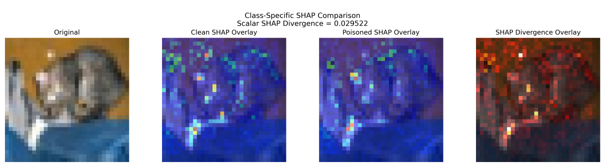

Fig. 8. SHAP Divergence Metric.

Qualitative SHAP analysis reveals attribution divergence between benign and poisoned models despite similar predictive accuracy. While predictions remain correct, internal feature importance patterns shift measurably, aligning with the instability hypothesis and providing interpretability support for structural distortion in parameter space.

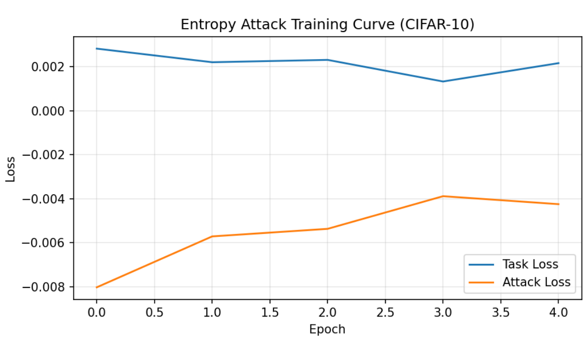

Fig. 9. Entropy-regularized adversarial fine-tuning on CIFAR-10.

Though the total task loss is unchanged across epochs, the entropy steadily reduces under adversarial fine-tuning, suggesting increasingly narrow predictive distributions as described by task loss without any discernible error on the task. The entropy distribution obtained seems to confirm the overconfident outputs that are clustered around low values of entropy, which are consistent with the sharp local minima and high sensitivity to parameter perturbations, thus supporting the theoretical connection between entropy minimization and instability amplification.

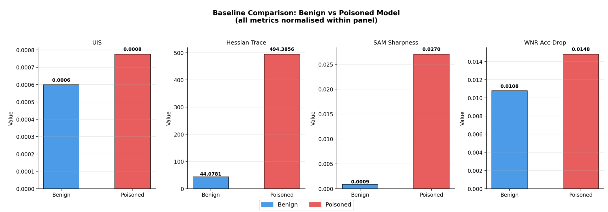

Fig. 11. A comparison of the structural instability metrics of benign and poisoned models.

UIS shows a significant difference between benign and poisoned updates, whereas Hessian trace, SAM sharpness, and weight-noise robustness showed less discrimination. This is evidence that forward-pass instability is effective in detecting structural distortions which are missing from the conventional sharpness proxies, thus confirming that UIS is orthogonal to and complementary to the conventional proxies, and is therefore a simple and appropriate detection statistic.

GradientSHAP attribution maps agree with predictions of labels but show that the spatial distribution of feature importance varies between benign and attacked models (attribution hotspots shift and redistribute across regions, consistent with the instability hypothesis).

The divergence of SHAPs is mostly concentrated around a mean of 1.8690, while the variance is not negligible. (Fig. 13), showing that the patterns of attribution are systematically modified, and not randomly. The increased variance in total attribution mass is present in the poisoned model (clean $\mu = 58.089$ vs poisoned $\mu = 51.652$) (Fig. 14) — showing that it is not a general increase in importance, but a redistribution instead. The 95th percentile CDF threshold of 2.7369 is useful for threshold-based attribution monitoring in deployment settings (Fig. 15). As shown in Fig. 16, attribution divergence is largest at $\lambda = 0.05$ (mean divergence $\approx$1.21), indicating that attribution displacement levels off as decision boundaries become more distinct. We observe the UIS–SHAP correlation ($r = -0.512$; see Fig. 17), which is in line with the

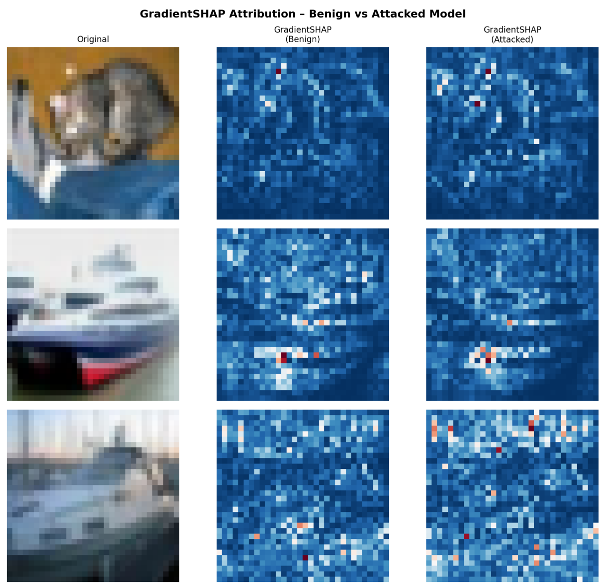

Fig. 12. A comparison of the gradientSHAP attributions for representative CIFAR-10 samples.

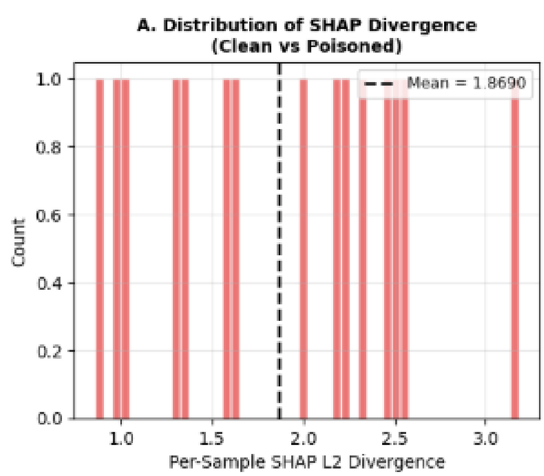

Fig. 13. Distribution of per-sample SHAP divergence.

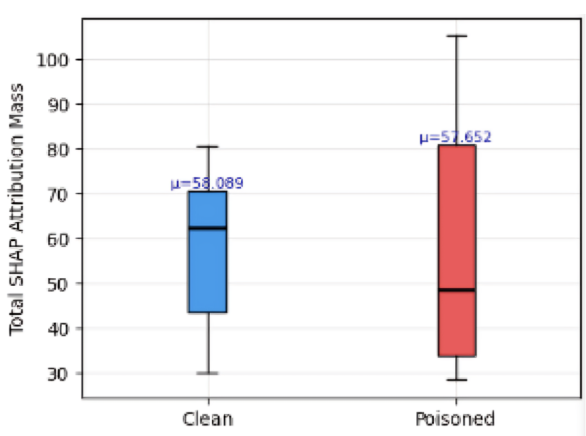

Fig. 14. SHAP mass distribution comparison per model.

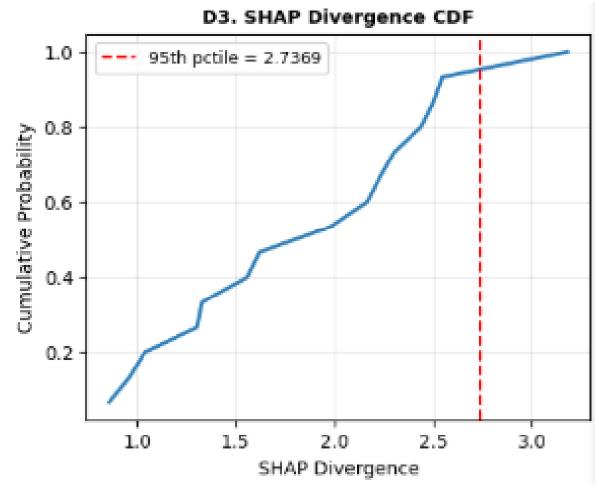

Fig. 15. SHAP Divergence CDF.

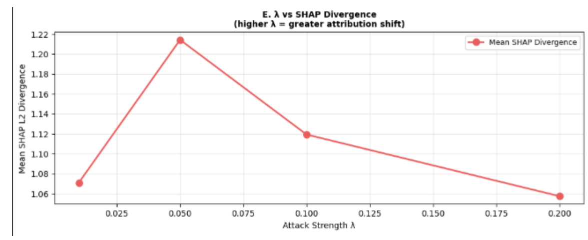

Fig. 16. Attribution shift under varying attack strength λ.

observed adversarial synchronization of both metrics, further reinforcing the coherent interpretation of structural distortion over both parameter and attribution space.

A. Operational Considerations

1) Threshold Selection: The UIS detection threshold is based on history of benign update statistics in an empirical manner. In particular, the threshold τ is set to:

$$\tau = \mu_b + 3\sigma_b \tag{10}$$

where μ_b and σ_b are the average and standard deviation of the values of UIS observed. The majority of the updates verified as benign were distributed across. If you choose this option, the benign UIS distribution will be generated. ($\mu = 0.00047$, $\sigma = 0.00015$) yields $\tau \approx 0.00092$, which remains more than two orders of magnitude below the minimum observed poisoned

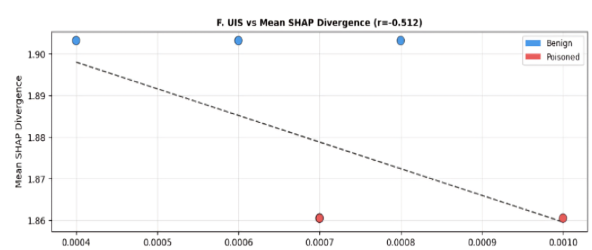

Fig. 17. Correlation between UIS and mean SHAP divergence ($r = -0.512$).

UIS of 0.07815. In practice, τ needs to be updated periodically due to new benign updates in the model registry.

TABLE V
UIS THRESHOLD CALIBRATION BENIGN REF D

Parameter	**Value**
Benign UIS mean (μ_b)	0.00047
Benign UIS std (σ_b)	0.00015
τ ($\mu + 2\sigma$)	0.00077
τ ($\mu + 3\sigma$) *[recommended]*	0.00092
Min. observed poisoned UIS	0.07815
Detection margin	$> 85\times$

2) Computational Overhead: There is only one forward pass for each perturbation trial, no gradient computing, and no change in architecture. The memory overhead is limited to an extra copy of the model parameters in the GPU memory, while the wall clock time is linearly scalable with $O(k \times N \times C_{fwd})$. Table VI summarizes the overhead comparison to estimation of hessian.

TABLE VI
UIS COMPUTATIONAL OVERHEAD VS HESSIAN TRACE
(RESNET-18, k=10)

Property	**UIS**	**Hessian Trace**
Backward passes	0	Required
Gradient storage	None	Required
Wall-clock (500 samples)	4.32s	$1.1\times$/batch
Time complexity	$O(k \times N)$	$O(k \times N \times \text{bwd})$

V. CONCLUSION

This paper shows that today's AI supply chains are structurally blind to what is happening: Stealthy adversarial model update poisoning is hard to detect using accuracy-based validation. We introduce the Update Instability Score (UIS), a simple metric that estimates the sensitivity of predictions to small perturbations in parameters that are known to affect predictions, without the need for Labeled data, retraining or gradient access. Experimental results on CIFAR-10 with cross-dataset validation on STL-10 demonstrate over two orders of magnitude separation between categories of benign and malicious updates, and non-overlapping confidence intervals for all tested conditions.

Future research will focus on transformer architectures, multimodal systems, large-scale foundations in production. Theoretical links between adversarial optimization, loss landscape geometry and parameter sensitivity will be further explored to gain insight into structural fragility mechanisms. As the AI landscape evolves in complexity, federated aggregation pipelines, model registries, and automated governance systems will continue to enhance the role UIS plays in their operation.

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